Answer Question – Model answers

INF10025 Data Management and Analytics · Test 3 revision (Weeks 1–8) · every SQL answer was run on a copy of the unit data and its result is shown

Level	Content	Questions
P – Pass	Weeks 1–6 core: concepts, Access criteria, ERD basics, one-table SQL, row counts, error codes	Q1–Q41
C – Credit	Joins, aggregates, GROUP BY / HAVING, constraints, identifiers, weak entities	Q42–Q60
D – Distinction	Week 8: subqueries, outer joins, set operators, ERD → schema, M:M, identifier chains	Q61–Q80
HD – High Distinction	Integrative scenarios, error prediction, design + SQL, code-segment ordering	Q81–Q90
Mock Test 3	12 mixed questions in Test-2 style (suggested time 30 min)	Section 5

How to use

Question numbers match Practice_Questions.pdf. “Verified result” boxes show the real output of the statement on the unit data (SQLite used for testing; where Oracle behaves differently – e.g. NULL ordering – the note says so).

ERD notation reminder

- **Strong entity** = single box; **weak entity** = double outline. Identifier attributes are underlined.
- **Crow’s foot** = MANY end; plain line = ONE end. “|” = **must**, “O” = **may**.
- An underlined relationship name means the entity at the MANY end borrows the identifier of the ONE end (weak entity).
- Foreign keys always go on the MANY side. A “must” at the ONE end → the FK is NOT NULL.
- A M:M relationship is replaced by a weak (intersection) entity with two M:1 relationships; no surrogate keys unless required.

Schema format used in the answers: **TABLE (col1, col2, …) PK (…) FK (…) → PARENT**.

Section 1 – Pass (P)

Q1. [Pass] Week 1 – data

Classify each item as structured, semi-structured or unstructured data: (a) a Student table with StudentID, Name, DateOfBirth; (b) an XML document; (c) a folder of customer e-mails and voice messages; (d) a NoSQL document store.

Model answer

(a) Structured – fixed pattern and data types for every row (RDBMS).
(b) Semi-structured – organised/arranged, but does not match the requirements of a relational database.
(c) Unstructured – no pre-defined format, content unpredictable (text, audio, links).
(d) Semi-structured (the lecture lists XML documents and NoSQL databases as examples).

Q2. [Pass] Week 1 – RDBMS

Which statement is NOT an advantage of an RDBMS compared with a spreadsheet?
a) Only the rows that match a search are retrieved
b) The data exists in one place so all users see a change immediately
c) Each user keeps a private copy of the data to work faster
d) Constraints and business rules are easier to enforce

Model answer

c) is not an advantage. In an RDBMS there is ONE copy of the data (on the database server); private copies are the spreadsheet problem (several versions of the “truth”).

Q3. [Pass] Week 1 – relational model

State the four properties of a two-dimensional table in the relational data model (Codd, 1970).

Model answer

1) Uniquely named columns (attributes). 2) A list of unnamed / unnumbered rows. 3) The order of the rows is irrelevant (shuffling keeps the meaning). 4) Only one value per cell – one cell for each attribute of a row.

Q4. [Pass] Week 1 – terminology

Complete the table of equivalent terms: Relation = ____ (Access) = ____ (Oracle); Attribute = ____ (Access) = ____ (Oracle); Tuple = ____ (Access) = ____ (Oracle).

Model answer

Relation = Table (Access) = Table (Oracle); Attribute = Field (Access) = Column (Oracle); Tuple = Record (Access) = Row (Oracle).

Q5. [Pass] Week 1 – primary key

A table has the column PhoneNo. Explain why PhoneNo is a poor choice of primary key by using the three properties of a primary key from Week 1.

Model answer

A primary key must (1) be a unique identifier – no duplicates (a family may share a phone number); (2) never be empty / NULL (not everybody has a phone number); (3) be static – never need to change (people change numbers). PhoneNo fails all three conditions, so use a real identifier such as StudentNo.

Q6. [Pass] Week 1 – Access views

Give TWO reasons why editing data in Datasheet View is dangerous, and name the Access view where field names and data types are defined.

Model answer

(1) A changed cell is saved permanently as soon as you move to another row – Access never asks “Do you want to save changes?”. (2) It is easy to delete one or many rows by mistake, and end-users would see data they should not see. Field names and data types are defined in Design View.

Q7. [Pass] Week 1 – filters vs queries

State two differences between a datasheet filter and a query.

Model answer

A filter is for one-off searches and its settings cannot be saved; a query is saved in the database, can be re-run many times and can be used as the source of another query (it behaves like a table).

Q8. [Pass] Week 1 – criteria grid

In the Access design grid the field RELYEAR has criteria 2013 on the “Criteria” row and 2016 on the “or” row. The field RATINGCODE has "M" on the Criteria row and "PG" on the “or” row. Write the equivalent SQL WHERE clause and say how many rows it returns from the MOVIE table.

Model answer

```
SELECT MOVIENO, TITLE, RELYEAR, RATINGCODE
FROM   MOVIE
WHERE  (RELYEAR = 2013 AND RATINGCODE = 'M')
OR     (RELYEAR = 2016 AND RATINGCODE = 'PG');
```

Verified result: 11 row(s) – first rows shown

MOVIENO	TITLE	RELYEAR	RATINGCODE
43074	Ghostbusters	2016	PG
49521	Man of Steel	2013	M
54138	Star Trek Into Darkness	2013	M
72190	World War Z	2013	M
101299	The Hunger Games: Catching Fire	2013	M
150117	I Give It a Year	2013	M

> **NOTE**
> Values on the same row are ANDed; different rows are ORed.

Q9. [Pass] Week 1 – impossible AND

A student types 2013 And 2016 in the Criteria cell of RELYEAR. How many rows are returned and why?

Model answer

0 rows. A single RELYEAR value can never equal both 2013 and 2016 – the AND condition is impossible. To get both years use Or (2013 Or 2016) or two rows in the grid.

Q10. [Pass] Week 1 – wildcards

Which titles match the Access criterion Like "the ????" ?
a) The Fall b) The Matrix c) Thelma and Louise d) the Hulk e) The Ring

Model answer

a), d) and e). “the ” (with the space) followed by exactly four characters; Access is not case sensitive. b) has 6 characters after “The ”; c) has no space after “The”.

Q11. [Pass] Week 1 – field names

Why does Access need square brackets in [movie title] but not in movieno? What advice does the lecture give about field names?

Model answer

Brackets are the only way to tell Access where a field name starts and ends when the name contains a space. The advice is to avoid spaces and use only A–Z, a–z and 0–9.

Q12. [Pass] Week 2 – FK

The MOVIE table has a column RATINGCODE and a table RATING has RATINGCODE as its primary key. (a) Which column is the foreign key? (b) Which table is the parent? (c) What kind of relationship is it? (d) What does “Enforce Referential Integrity” stop you doing?

Model answer

(a) MOVIE.RATINGCODE. (b) RATING (the table with the primary key, the ONE end). (c) Many-to-One (many movies, one rating). (d) It stops you entering a RATINGCODE in MOVIE (e.g. ‘X’) that does not exist in RATING, and stops you deleting a rating that is still used by movies.

Q13. [Pass] Week 2 – Cartesian product

MOVIE has 395 rows and RATING has 5 rows. If the two tables are not joined in a query, how many rows come back? Write the query that shows it.

Model answer

```
SELECT COUNT(*) FROM MOVIE, RATING;
```

Verified result: 1 row(s)

COUNT(*)
1975

> **NOTE**
> 395 × 5 = 1975 rows – every movie is matched with every rating (a Cartesian product).

Q14. [Pass] Week 2 – totals

Write an SQL statement equivalent to an Access Totals query that shows each rating code and the number of movies with that code (heading “Total”).

Model answer

```
SELECT RATINGCODE, COUNT(*) AS "Total"
FROM   MOVIE
GROUP BY RATINGCODE;
```

Verified result: 4 row(s)

RATINGCODE	Total
G	16
M	181
MA	78
PG	120

Q15. [Pass] Week 2 – indexes

(a) What two items does an index store? (b) Name one advantage and one disadvantage of indexing every column. (c) Which index does Access create automatically?

Model answer

(a) The index value (key) and the location of the record (RowLoc). (b) Advantage: much faster searching/sorting/joins and reports (fewer disk I/Os). Disadvantage: every INSERT/UPDATE/DELETE must also update every index, so changes become slow (disk space is only a minor cost). (c) The primary key index (no duplicates).

Q16. [Pass] Week 2 – indexes

A table has 100,000 rows and no index on Surname. On average, about how many rows must be read sequentially to find a surname? a) 1 b) 50,000 c) 100,000 d) 2

Model answer

b) About 50,000 (on average half the table). With an index: one read for the index, binary search, one read for the row.

Q17. [Pass] Week 3 – parameters

Write the criterion you type in the RELYEAR cell so that Access asks the user for a year when the query is run, and name two data types you can choose for a parameter.

Model answer

= [Find Year] (any name inside square brackets). Data types: ShortText, Date, Integer, Decimal.

Q18. [Pass] Week 3 – action queries

Which of these can be undone with Ctrl+Z after running? Update query, Delete query, Make Table query, Select query. What should you do before running an action query?

Model answer

None of the action queries (Update, Delete, Make Table, Append) can be undone with Ctrl+Z; a Select query changes nothing. Before running an action query make a backup/copy of the table (and test with a Select first).

Q19. [Pass] Week 3 – import

Explain the difference between importing and linking external data in Access.

Model answer

Import makes a copy of the data inside Access tables. Link keeps ONE version of the data: it is maintained by the original application and Access has only a read-only view of it.

Q20. [Pass] Week 3 – Excel dates

How does Excel store the date 2 January 1900, and why may imported Excel dates look like numbers?

Model answer

Excel stores dates as serial numbers – the count of days since 1 January 1900 (the fraction stores the time), so 2 Jan 1900 is 2. After import the values may look like numbers, but Access interprets them as dates.

Q21. [Pass] Week 4 – Power BI

(a) What does the Relationships window show? (b) Which cross-filter direction should be selected for every relationship? (c) What is the difference between Power BI Desktop and the Power BI web app?

Model answer

(a) The relationships between the imported tables (transferred from Access; more can be added). (b) Both (two-way arrows). (c) Desktop is the Windows application where data is imported/manipulated and reports/measures are created; the web app is the cloud service where deployed reports/dashboards are used by end-users and analysts.

Q22. [Pass] Week 4 – functions

Title = "The Matrix". Give the result of: (a) Left([title],3) (b) Right([title],4) (c) Mid([title],5,3) (d) Ucase([title]) in Access, (e) upper([title]) in Power BI.

Model answer

(a) The (b) trix (c) Mat (d) THE MATRIX (e) THE MATRIX. Positions start at 1.

Q23. [Pass] Week 5 – ERD concepts

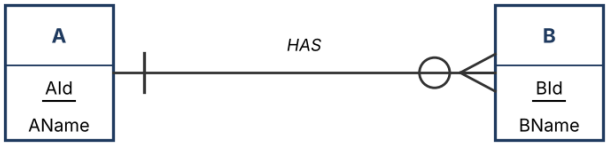
Define: entity, attribute, identifier, business rule. Why is an ERD “a plan” and not a database?

Model answer

Entity – a real-world thing about which data is stored (person, object, event). Attribute – a property of an entity. Identifier – attribute(s) that uniquely identify each instance (no duplicates, never null) – underlined, = primary key. Business rule – a rule or policy of the organisation that governs the data (e.g. a subject must have one convenor). An ERD stores no data; it is drawn before the database is built to decide which tables are needed.

Q24. [Pass] Week 5 – reading ERD

For the ERD below read the relationship in both directions (each sentence must start with the word ONE), and state which table receives the foreign key.

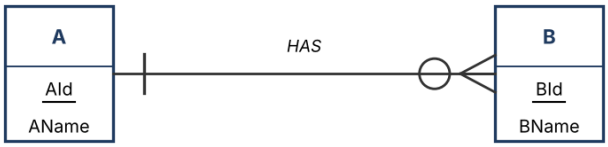


Model answer

ONE A may have MANY B (reading left to right: HAS). ONE B must be had by ONE A (right to left). The foreign key AId is added to table B (the MANY end).

Q25. [Pass] Week 5 – schema

Convert the ERD above into a relational schema. Show primary and foreign keys.



Model answer

```
A (AId, AName)
    PK (AId)
B (BId, BName, AId)
    PK (BId)
    FK (AId) references A
```

Q26. [Pass] Week 5 – identifier rules

List the four rules for an identifier in an ERD.

Model answer

Every entity must have an identifier; every instance must have a unique identifier (no duplicates); the value can never be empty/null; identifiers are underlined on the ERD.

Q27. [Pass] Week 5 – NULL

LECTURER(LecId, LecName, Age): lecturer 207 is 37, lecturer 771 Karen Fry has Age = NULL. Does Karen appear in SELECT * FROM lecturer WHERE age > 37? In WHERE age <= 37? In WHERE age IS NULL? Explain.

Model answer

No / No / Yes. Comparing NULL with a value gives UNKNOWN (neither TRUE nor FALSE), so the row is excluded by both > 37 and <= 37. Only IS NULL finds her. NULL is not zero and not a space.

Q28. [Pass] Week 5 – DDL

Write the CREATE TABLE statements (Oracle) for LECTURER(LecId, LecName, Age) and SUBJECT(SubjectCode, Title, CreditPoints, LecId) with the primary keys, the foreign key, and a constraint that every subject MUST have a convenor.

Model answer

```
CREATE TABLE Lecturer (
  LecId   NUMBER,
  LecName VARCHAR(50),
  Age     NUMBER,
  PRIMARY KEY (LecId) );

CREATE TABLE Subject (
  SubjectCode  VARCHAR(10),
  Title        VARCHAR(100),
  CreditPoints NUMBER,
  LecId        NUMBER NOT NULL,
  PRIMARY KEY (SubjectCode),
  FOREIGN KEY (LecId) REFERENCES Lecturer );
-- Lecturer must be created first (parent before child).
```

Q29. [Pass] Week 5 – INSERT

Write INSERT statements: lecturer 974 Robert Tipping whose age is unknown, and the subject INF21009 “Advanced Facebook” (12.5 credit points) convened by lecturer 345. Which common mistake gives ORA-00911?

Model answer

```
INSERT INTO Lecturer (LecId, LecName, Age) VALUES (974, 'Robert Tipping', Null);
INSERT INTO Subject (SubjectCode, Title, CreditPoints, LecId)
VALUES ('INF21009', 'Advanced Facebook', 12.5, 345);
```

> **NOTE**
> Writing ‘Null’ (quoted) would store the text Null. ORA-00911 invalid character is caused by smart quotes ‘ ’ pasted from Word – use Notepad / Notepad++ and straight single quotes.

Q30. [Pass] Week 5 – spot the error

Say whether each statement is valid in Oracle and fix any that are not (table EMPLOYEE(EmpId, EmpName, Gender)):
1 SELECT EmpName, Gender
2 SELECT * FROM Employee ORDER BY EmpName
3 SELECT EmpName FROM Employee WHERE Gender Is M
4 SELECT EmpId, EmpName, Gender WHERE Gender = 'Female' ORDER BY EmpId ASC
5 SELECT LecId LecName Age FROM LECTURER

Model answer

1 Invalid – no FROM clause.
2 Valid.
3 Invalid – text must be quoted and = is needed: WHERE Gender = 'M'.
4 Invalid – FROM clause missing.

5 Invalid – commas missing between the column names (SELECT LecId, LecName, Age FROM LECTURER).

Q31. [Pass] Week 5 – UPDATE/DELETE

Explain what happens if you run UPDATE employee SET salary = salary + 5000; (no WHERE). Describe the safe way to write a DELETE.

Model answer

Every row is updated – all salaries increase by 5000 and there is usually no undo. For DELETE first write the SELECT with the WHERE clause, check the rows returned, then change SELECT * FROM … into DELETE FROM … with the same WHERE.

Q32. [Pass] Week 5 – ORDER BY

Write a single statement listing LecName and Age of all lecturers in ascending Age sequence and, within the same age, in descending LecName sequence. Then write another using the column number instead of names.

Model answer

```
SELECT LecName, Age
FROM   Lecturer
ORDER BY Age ASC, LecName DESC;

SELECT LecName, Age
FROM   Lecturer
ORDER BY 2, 1 DESC;
```

Verified result (statement 1): 4 row(s)

LecName	Age
Jane Pitt	26
Carol Kent	34
John Smith	37
Lee Wong	45

Verified result (statement 2): 4 row(s)

LecName	Age
Jane Pitt	26
Carol Kent	34
John Smith	37
Lee Wong	45

Q33. [Pass] Week 6 – row counts

Use the CUSTOMER table (custno, custname, typecode, sales, location) supplied with the Test-2 style data (see Section 0.1). How many rows does each statement return?

Model answer

Verified results:

Part	Result
a	3
b	0
c	0
d	4
e	2
f	4
g	3

Part	Result
h	0
i	2
j	6

> **NOTE**
> Oracle compares text case-sensitively; AND is evaluated before OR; = NULL is never TRUE.

Q34. [Pass] Week 6 – ORDER BY

Which customer name is listed second by: SELECT custname FROM customer WHERE typecode > 1 ORDER BY custname; ?

Model answer

```
SELECT custname
FROM   customer
WHERE  typecode > 1
ORDER BY custname;
```

Verified result: 6 row(s)

custname
Bob
Dave
Helen
Nick
Sue
Xena

> **NOTE**
> Result order: Bob, Dave, Helen, Nick, Sue, Xena – the second name is Dave.

Q35. [Pass] Week 6 – write SQL

Write a single SQL statement to list all customer names and sales amounts in descending sales sequence.

Model answer

```
SELECT custname, sales
FROM   customer
ORDER BY sales DESC;
```

Verified result: 9 row(s) – first rows shown

custname	sales
Sue	700.0
Jenny	525.0
Wendy	300.0
Helen	200.0
Dave	95.0
Linda	25.0

> **NOTE**
> Oracle sorts NULL first in DESC order (NULLs are treated as the largest value).

Q36. [Pass] Week 6 – write SQL

List the names of customers whose sales are between 25 and 100 inclusive OR between 500 and 900 inclusive.

Model answer

```
SELECT custname
FROM   customer
WHERE  (sales >= 25 AND sales <= 100)
OR     (sales >= 500 AND sales <= 900);
```

Verified result: 4 row(s)

custname
Sue
Dave
Jenny
Linda

> **NOTE**
> Linda 25, Dave 95, Jenny 525, Sue 700 (BETWEEN limits are inclusive).

Q37. [Pass] Week 6 – write SQL

List the customer names that contain the letter N (upper or lower case) and the customer names that have a NULL sales amount – as two separate statements.

Model answer

```
SELECT custname FROM customer WHERE UPPER(custname) LIKE '%N%';
SELECT custname FROM customer WHERE sales IS NULL;
```

Verified result (statement 1): 6 row(s)

custname
Wendy
Nick
Jenny
Helen
Linda
Xena

Verified result (statement 2): 2 row(s)

custname
Nick
Xena

Q38. [Pass] Week 6 – write SQL

List each location only once (no duplicates) in alphabetical order. Then list customers whose typecode is 1 or 3 using IN.

Model answer

```
SELECT DISTINCT location FROM customer ORDER BY location;
SELECT custname, typecode FROM customer WHERE typecode IN (1, 3);
```

Verified result (statement 1): 3 row(s)

location
Haw
Kew
Richmond

Verified result (statement 2): 6 row(s)

custname	typecode
Wendy	1
Sue	3
Jenny	1

custname	typecode
Helen	3
Linda	1
Xena	3

Q39. [Pass] Week 6 – UPPER trap

Why does SELECT * FROM customer WHERE UPPER(location) = 'Haw' return 0 rows? Correct it in two ways.

Model answer

```
SELECT * FROM customer WHERE UPPER(location) = 'HAW';
SELECT * FROM customer WHERE location = 'Haw';
```

Verified result (statement 1): 4 row(s)

custno	custname	typecode	sales	location
21	Wendy	1	300.0	Haw
24	Sue	3	700.0	Haw
25	Jenny	1	525.0	Haw
23	Helen	3	200.0	Haw

Verified result (statement 2): 4 row(s)

custno	custname	typecode	sales	location
21	Wendy	1	300.0	Haw
24	Sue	3	700.0	Haw
25	Jenny	1	525.0	Haw
23	Helen	3	200.0	Haw

NOTE
UPPER(location) can never equal a mixed-case literal; compare upper with upper, or compare the original with the exact stored text.

Q40. [Pass] Week 6 – BETWEEN/LIKE

Rewrite WHERE age BETWEEN 30 AND 40 using AND, and WHERE age IN (24,26,28) using OR. Which letters does LIKE 'J%' require?

Model answer

age >= 30 AND age <= 40 (BETWEEN is inclusive); age = 24 OR age = 26 OR age = 28; the value must START with J (% stands for any number of characters).

Q41. [Pass] Week 6 – errors

Match the Oracle error to the situation: (1) inserting a subject whose LecId does not exist in LECTURER; (2) deleting a lecturer who still convenes subjects; (3) inserting NULL into a NOT NULL column; (4) smart quotes; (5) SELECT relyear, COUNT(*) FROM movie;

Model answer

(1) ORA-02291 parent key not found. (2) ORA-02292 child record found. (3) ORA-01400 cannot insert NULL. (4) ORA-00911 invalid character. (5) ORA-00937 not a single-group group function.

Section 2 – Credit (C)

Q42. [Credit] Week 6 – join

Write one SQL statement that lists each customer name with the Title of its customer type (tables CUSTOMER and TYPE). How many rows are returned?

Model answer

```
SELECT c.custname, t.title
FROM   customer c, type t
WHERE  c.typecode = t.typecode;
```

Verified result: 9 row(s) – first rows shown

custname	Title
Wendy	Under 20
Sue	20 plus
Nick	Under 20
Dave	20 plus
Jenny	Under 20
Bob	20 plus

> **NOTE**
> 9 rows – every customer has a valid type code; the join condition is the PK/FK pair.

Q43. [Credit] Week 6 – join (3 tables)

List customer name, type title and grade description for customers whose grade description is "Women and Children", in alphabetical order of customer name.

Model answer

```
SELECT c.custname, t.title, g.description
FROM   customer c, type t, grade g
WHERE  c.typecode = t.typecode
AND    t.gradeid  = g.gradeid
AND    g.description = 'Women and Children'
ORDER BY c.custname;
```

Verified result: 6 row(s)

custname	Title	description
Helen	20 plus	Women and Children
Jenny	Under 20	Women and Children
Linda	Under 20	Women and Children
Sue	20 plus	Women and Children
Wendy	Under 20	Women and Children
Xena	20 plus	Women and Children

> **NOTE**
> N tables need N-1 join conditions (here 2). Customer types 1 and 3 have grade WC.

Q44. [Credit] Week 6 – join syntax

Rewrite the previous question using the JOIN … ON syntax. Which join conditions belong in ON and which in WHERE?

Model answer

```
SELECT c.custname, t.title, g.description
FROM   customer c
JOIN   type  t ON c.typecode = t.typecode
JOIN   grade g ON t.gradeid  = g.gradeid
WHERE  g.description = 'Women and Children'
```

```
ORDER BY c.custname;
```

Verified result: 6 row(s)

custname	Title	description
Helen	20 plus	Women and Children
Jenny	Under 20	Women and Children
Linda	Under 20	Women and Children
Sue	20 plus	Women and Children
Wendy	Under 20	Women and Children
Xena	20 plus	Women and Children

NOTE
Join conditions go in ON; row filters stay in WHERE.

Q45. [Credit] Week 6 – join (subjects)

List the title of each subject together with the name of the lecturer who convenes it. How many rows are returned, and which subject is missing from the result? Why?

Model answer

```
SELECT s.title, l.lecname
FROM   subject s, lecturer l
WHERE  s.lecid = l.lecid;
```

Verified result: 4 row(s)

Title	LecName
Intro to Web	John Smith
EBIS	Carol Kent
Honours Project	John Smith
Media Thesis A	Jane Pitt

NOTE
4 rows. INF21009 “Advanced Facebook” is missing because its LecId is NULL – a NULL never matches in an inner join.

Q46. [Credit] Week 6 – join + ORDER

List lecturer name, office number and year painted for every lecturer (tables LECTURER, OFFICE). Which lecturer is missing and why?

Model answer

```
SELECT l.lecname, o.officeno, o.yearpainted
FROM   lecturer l, office o
WHERE  l.officeno = o.officeno;
```

Verified result: 3 row(s)

LecName	OfficeNo	YearPainted
John Smith	EN710	2004
Jane Pitt	EN505	2011
Carol Kent	EN710	2004

NOTE
3 rows – Lee Wong has no office (NULL foreign key), so the inner join drops him.

Q47. [Credit] Week 6 – self reference

Show every pair of different players (Pid1, Pid2) who play in the same campaign, listing each pair once only.

Model answer

```
SELECT p1.pname, p2.pname
FROM   player p1, player p2
WHERE  p1.campaigncode = p2.campaigncode
AND    p1.pid < p2.pid;
```

Verified result: 2 row(s)

PName	PName
Ali Aien	Sheelan Aien
Robert Tipping	Lynda Rogers

> **NOTE**
> Use < (not <>) to remove the mirrored pair. Answer: 2 pairs (Ali/Sheelan, Robert/Lynda).

Q48. [Credit] Week 7 – aggregates

What does each statement return for the EMPLOYEE table? (a) COUNT(*) (b) COUNT(salary) (c) AVG(salary) (d) SUM(salary)/COUNT(*)

Model answer

Verified results:

Part	Result
a	7
b	6
c	62333.33
d	53428.57

> **NOTE**
> Aggregate functions ignore NULL salaries (Eve). COUNT(*) counts rows; AVG divides by the 6 non-null values, so (c) ≠ (d).

Q49. [Credit] Week 7 – GROUP BY

Write the statement that shows each DeptNo and the average salary of that department, rounded to 0 decimals, sorted by DeptNo. How many groups appear? What happens to Gina?

Model answer

```
SELECT deptno, ROUND(AVG(salary),0) AS avg_sal
FROM   employee
GROUP BY deptno
ORDER BY deptno;
```

Verified result: 4 row(s)

DeptNo	avg_sal
NULL	60000.0
10	50000.0
20	84500.0
30	45000.0

> **NOTE**
> 4 groups: 10, 20, 30 and one group with deptno NULL (Gina) – NULLs are grouped together. Oracle lists NULL last in ascending order.

Q50. [Credit] Week 7 – HAVING

List the departments that have more than one employee, with the employee count.

Model answer

```
SELECT deptno, COUNT(*) AS cnt
FROM   employee
GROUP BY deptno
```

```
HAVING COUNT(*) > 1;
```

Verified result: 2 row(s)

DeptNo	cnt
10	2
20	3

> **NOTE**
> HAVING filters groups after aggregation; WHERE filters rows before it. Result: dept 10 (2) and dept 20 (3).

Q51. [Credit] Week 7 – WHERE vs HAVING

Explain why this statement fails and rewrite it so it lists the departments (deptno) whose average salary is above 50000: SELECT deptno FROM employee WHERE AVG(salary) > 50000 GROUP BY deptno;

Model answer

```
SELECT deptno
FROM   employee
GROUP BY deptno
HAVING AVG(salary) > 50000;
```

Verified result: 2 row(s)

DeptNo
NULL
20

> **NOTE**
> Aggregates are not allowed in WHERE (ORA-00934). Result: dept 20 (avg 84,500) and the NULL-dept group (60,000), dept 10 is 50,000 – not greater.

Q52. [Credit] Week 7 – movies

How many movies are in each rating? Show RATINGCODE and the count, highest count first.

Model answer

```
SELECT ratingcode, COUNT(*) AS total
FROM   movie
GROUP BY ratingcode
ORDER BY total DESC;
```

Verified result: 4 row(s)

RATINGCODE	total
M	181
PG	120
MA	78
G	16

Q53. [Credit] Week 7 – movies

Show the year, number of movies and average runtime of the years that have at least 15 movies.

Model answer

```
SELECT relyear, COUNT(*) AS n, AVG(runtime) AS avg_rt
FROM   movie
GROUP BY relyear
HAVING COUNT(*) >= 15
ORDER BY relyear;
```

Verified result: 8 row(s) – first rows shown

RELYEAR	n	avg_rt
2004	19	114.89
2005	15	127.0
2011	17	115.29
2012	26	125.81
2013	17	121.18
2014	29	120.72

NOTE
8 years: 2004, 2005, 2011–2014, 2016, 2018.

Q54. [Credit] Week 7 – join + aggregate

List each rating NAME (table RATING) and the number of movies in it.

Model answer

```
SELECT r.ratingname, COUNT(*) AS total
FROM   movie m, rating r
WHERE  m.ratingcode = r.ratingcode
GROUP BY r.ratingname;
```

Verified result: 4 row(s)

RATINGNAME	total
General	16
Mature (15+)	181
Mature Accompanied (15+)	78
Parental Guidance	120

NOTE
Rating R has no movies, so it does not appear (inner join) – see the outer-join questions at level D.

Q55. [Credit] Week 7 – DDL constraints

Write ALTER/CREATE statements for: a table PRODUCT(ProdNo number PK, Descr varchar(30) NOT NULL, Price number CHECK Price > 0, Colour varchar(10) CHECK Colour IN ('Red','Blue'), Code varchar(5) UNIQUE).

Model answer

```
CREATE TABLE Product (
  ProdNo NUMBER PRIMARY KEY,
  Descr  VARCHAR(30) NOT NULL,
  Price  NUMBER CHECK (Price > 0),
  Colour VARCHAR(10) CHECK (Colour IN ('Red','Blue')),
  Code   VARCHAR(5) UNIQUE );
```

Q56. [Credit] Week 7 – CHECK/UNIQUE

Table T has UNIQUE(Code) and CHECK(Price > 0). For each insert say whether it succeeds: (a) (1,'A1', 5) (b) (2,'A1', 9) (c) (3, NULL, 7) (d) (4, NULL, 8) (e) (5,'B2', NULL) (f) (6,'C3', 0). Name the error codes.

Model answer

(a) succeeds. (b) fails ORA-00001 unique constraint (duplicate A1). (c) succeeds – UNIQUE allows NULL. (d) succeeds – a second NULL does not count as duplicate. (e) succeeds – a CHECK passes when the condition is TRUE or NULL/UNKNOWN. (f) fails ORA-02290 check constraint violated (0 > 0 is FALSE).

Q57. [Credit] Week 6 – dates

Write a query listing employees hired after 31 December 2018 using TO_DATE, and the Access equivalent of the date literal.

Model answer

```
SELECT empname, hiredate FROM employee
WHERE  hiredate > TO_DATE('31/12/2018','dd/mm/yyyy');
-- Access: WHERE hiredate > #31/12/2018#   (or #12/31/2018# depending on region settings; Access is not
    case sensitive)
```

Q58. [Credit] Week 5 – identifier test

The table below holds sample data for entity ENROLMENT. Which attribute(s) form a valid identifier? Test the candidates: (StuId), (SubCode), (StuId, SubCode), (StuId, SubCode, EnrolDate).

SubCode | StuId | EnrolDate | FinalGrade
INF10002 | 1122334 | 28-Feb-2014 | HD
INF20014 | 7777700 | 28-Feb-2014 | P
INF10002 | 4444555 | 28-Feb-2014 | C
INF30004 | 4444555 | 15-Jul-2014 | P
INF10002 | 7777700 | 15-Jul-2014 | HD
INF20014 | 1122334 | 27-Feb-2015 | D

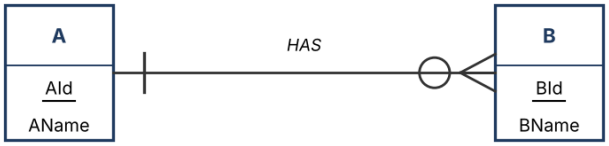
Model answer

StuId alone is not unique (1122334 repeats); SubCode alone is not unique (INF10002 repeats). (StuId, SubCode) is unique in the sample, but the business rule that a student may retake a subject in a later year means the safe identifier is (StuId, SubCode, EnrolDate) – the minimal set that is unique under the rule. FinalGrade can be NULL (not yet graded) so it cannot be part of the identifier.

> **NOTE**
> The sample data cannot prove a rule – the business rule decides.

Q59. [Credit] Week 5 – 1:M ERD

Read the ERD below. (a) What is the relationship in words (both directions)? (b) Is B a weak entity? (c) Write the schema.

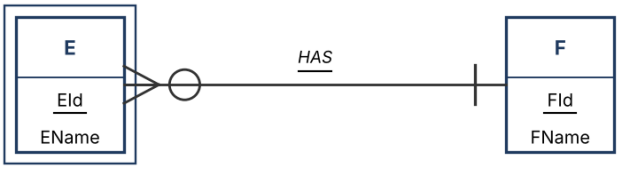


Model answer

(a) ONE A may have MANY B (the circle at the B end = may) / ONE B must belong to ONE A (the bar at the A end = must). (b) No – the relationship name is not underlined, so B has its own identifier BId. (c) A(AId, …) B(BId, …, AId FK).

Q60. [Credit] Week 5 – weak entity

In the ERD below the relationship between E and F is underlined (E has a double outline). State the identifier of E in the relational schema and write the schema.



Model answer

```
F (FId, FName)
    PK (FId)
E (FId, EId, EName)
    PK (FId, EId)
    FK (FId) references F
```

NOTE
The underlined relationship means E borrows the identifier of F: PK of E = FK from D + own partial identifier EId.

Section 3 – Distinction (D)

Q61. [Distinction] Week 8 – subquery

List the names of employees who earn more than the average salary of all employees.

Model answer

```
SELECT empname, salary
FROM   employee
WHERE  salary > (SELECT AVG(salary) FROM employee);
```

Verified result: 2 row(s)

EmpName	Salary
Chloe	91000.0
Dan	78000.0

> **NOTE**
> Average = 62,333 (NULL ignored): Chloe 91,000 and Dan 78,000. A single-value subquery can use >.

Q62. [Distinction] Week 8 – subquery IN

List the departments (DeptName) that have at least one employee earning over 60000.

Model answer

```
SELECT deptname
FROM   dept
WHERE  deptno IN (SELECT deptno FROM employee WHERE salary > 60000);
```

Verified result: 1 row(s)

DeptName
IT

> **NOTE**
> Only IT (Chloe, Dan). Gina (60,000 / NULL dept) does not qualify.

Q63. [Distinction] Week 8 – NOT IN trap

Departments with NO employees. (a) Write it with NOT IN. (b) Explain what happens if the subquery returns a NULL (Gina has NULL DeptNo). (c) Give a safe version.

Model answer

```
SELECT deptname FROM dept
WHERE  deptno NOT IN (SELECT deptno FROM employee);

SELECT deptname FROM dept
WHERE  deptno NOT IN (SELECT deptno FROM employee WHERE deptno IS NOT NULL);

SELECT d.deptname FROM dept d
WHERE  NOT EXISTS (SELECT 1 FROM employee e WHERE e.deptno = d.deptno);
```

Verified result (statement 1): 0 row(s)

DeptName

Verified result (statement 2): 1 row(s)

DeptName
Legal

Verified result (statement 3): 1 row(s)

DeptName
Legal

NOTE
The first statement returns 0 rows: x NOT IN (10,20,30,NULL) is UNKNOWN for every x not in the list. Safe versions return Legal (dept 40).

Q64. [Distinction] Week 8 – outer join

Show every rating name and the number of movies in it, INCLUDING ratings with no movies (count = 0).

Model answer

```
SELECT r.ratingname, COUNT(m.movieno) AS total
FROM   rating r LEFT OUTER JOIN movie m
       ON r.ratingcode = m.ratingcode
GROUP BY r.ratingname;
```

Verified result: 5 row(s)

RATINGNAME	total
General	16
Mature (15+)	181
Mature Accompanied (15+)	78
Parental Guidance	120
Restricted (18+)	0

NOTE
Count a column of the optional side (m.movieno), never COUNT(*), otherwise the unmatched row counts as 1.

Q65. [Distinction] Week 8 – outer join

List every subject title with its convenor name; subjects without a convenor must still appear (show NULL).

Model answer

```
SELECT s.title, l.lecname
FROM   subject s LEFT OUTER JOIN lecturer l
       ON s.lecid = l.lecid;
```

Verified result: 5 row(s)

Title	LecName
Intro to Web	John Smith
EBIS	Carol Kent
Honours Project	John Smith
Media Thesis A	Jane Pitt
Advanced Facebook	NULL

Q66. [Distinction] Week 8 – outer join

List all lecturers with the year their office was painted, including lecturers who have no office.

Model answer

```
SELECT l.lecname, o.yearpainted
FROM   lecturer l LEFT OUTER JOIN office o
       ON l.officeno = o.officeno;
```

Verified result: 4 row(s)

LecName	YearPainted
John Smith	2004
Jane Pitt	2011
Carol Kent	2004

LecName	YearPainted
Lee Wong	NULL

Q67. [Distinction] Week 8 – anti-join

List the players who are NOT in "Campaign Three" without using NOT IN (use an outer join and a NULL test).

Model answer

```
SELECT p.pname
FROM   player p LEFT OUTER JOIN campaign c
       ON p.campaigncode = c.campaigncode AND c.campaigntitle = 'Campaign Three'
WHERE  c.campaigncode IS NULL;
```

Verified result: 4 row(s)

PName
Ali Aien
Robert Tipping
Lynda Rogers
Sheelan Aien

> **NOTE**
> Put the filter in the ON clause so the outer join keeps unmatched players; then keep the rows with no match.

Q68. [Distinction] Week 8 – set operators

Table UNI_STUDENT has Amanda Allen, Bob Black, Carol Chen, Dave Dole (4 rows) and UNI_TUTOR has Carol Chen, Ed Ellis, Fred Foley (3 rows). How many rows does each return: UNION, UNION ALL, INTERSECT, MINUS (student MINUS tutor)?

Model answer

Verified results:

Part	Result
UNION	6
UNION ALL	7
INTERSECT	1
MINUS	3

> **NOTE**
> Set operators need the same number of columns with compatible types; UNION removes duplicates, UNION ALL does not. Answers: 6, 7, 1, 3.

Q69. [Distinction] Week 8 – MINUS in Access

Access has no INTERSECT/MINUS. Rewrite “movies that are rated G but not in colour code 1” style problems (A MINUS B) in a form Access accepts.

Model answer

Use NOT IN / NOT EXISTS or a LEFT JOIN … WHERE b.key IS NULL. INTERSECT is rewritten with IN / EXISTS or an inner join.

Q70. [Distinction] Week 8 – correlated subquery

List employees who earn more than the average salary of THEIR OWN department.

Model answer

```
SELECT e.empname, e.deptno, e.salary
FROM   employee e
WHERE  e.salary > (SELECT AVG(salary) FROM employee e2 WHERE e2.deptno = e.deptno);
```

Verified result: 2 row(s)

EmpName	DeptNo	Salary
Anna	10	52000.0
Chloe	20	91000.0

> **NOTE**
> Anna (52,000 > 50,000) and Chloe (91,000 > 84,500). Dan is below 84,500; Gina has a NULL department so the inner AVG is NULL.

Q71. [Distinction] Week 8 – top rows

Which rating has the highest average runtime and what is it? Write it with a subquery in HAVING.

Model answer

```
SELECT ratingcode, AVG(runtime) AS avg_rt
FROM   movie
GROUP BY ratingcode
HAVING AVG(runtime) = (SELECT MAX(AVG(runtime)) FROM movie GROUP BY ratingcode);
```

> **NOTE**
> M (124.04). Nested aggregate MAX(AVG()) is valid in Oracle.

Q72. [Distinction] Week 8 – outer join counts

Which actors never appear in CASTING? Write it with a LEFT JOIN and a NULL test, and give the row count.

Model answer

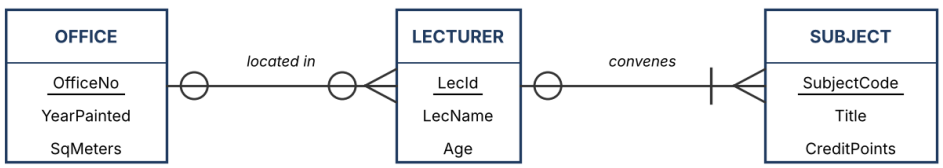
```
SELECT a.actorno, a.fullname
FROM   actor a LEFT OUTER JOIN casting c ON a.actorno = c.actorno
WHERE  c.castid IS NULL;
```

Verified result: 3 row(s)

ActorNo	FullName
7	Andrew Stanton
8293	Marion Cotillard
90633	Gal Gadot

Q73. [Distinction] Week 8 – ERD to schema

Convert the following ERD (OFFICE – LECTURER – SUBJECT) to a relational schema. Where do the foreign keys go? Which foreign keys may be NULL?



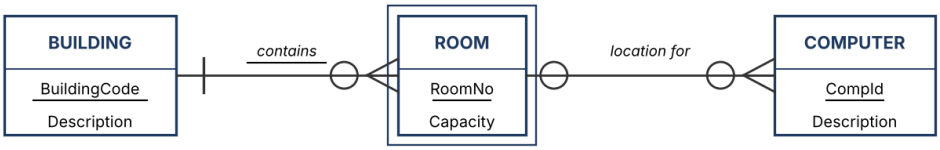
Model answer

```
OFFICE (OfficeNo, YearPainted, SqMeters)
    PK (OfficeNo)
LECTURER (LecId, LecName, Age, OfficeNo)
    PK (LecId)
    FK (OfficeNo) references OFFICE      -- may be NULL (lecturer may have no office)
SUBJECT (SubjectCode, Title, CreditPoints, LecId)
    PK (SubjectCode)
    FK (LecId) references LECTURER       -- NOT NULL (subject MUST have a convenor)
```

NOTE
FK always goes on the MANY side; a “must” at the ONE end → NOT NULL FK; “may” → NULL allowed.

Q74. [Distinction] Week 8 – ERD to schema

Convert the ERD below (BUILDING – ROOM – COMPUTER; ROOM is weak) to a schema. What is the primary key of ROOM and of COMPUTER? Write the foreign key of COMPUTER.



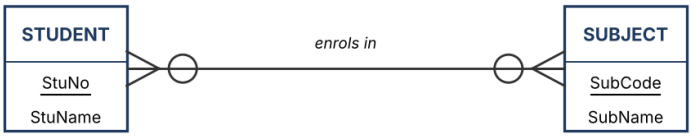
Model answer

```
BUILDING (BuildingCode, Description)
    PK (BuildingCode)
ROOM (BuildingCode, RoomNo, Capacity)
    PK (BuildingCode, RoomNo)
    FK (BuildingCode) references BUILDING
COMPUTER (CompId, Description, BuildingCode, RoomNo)
    PK (CompId)
    FK (BuildingCode, RoomNo) references ROOM   -- composite FK, both columns may be NULL together
```

NOTE
The composite key of the weak entity ROOM is copied as a composite FK into COMPUTER.

Q75. [Distinction] Week 8 – M:M

The ERD below shows a many-to-many relationship. (a) Why can it not be implemented directly? (b) Redraw it as an ERD with a new entity and give the schema, assuming enrolment date and final grade must be recorded.



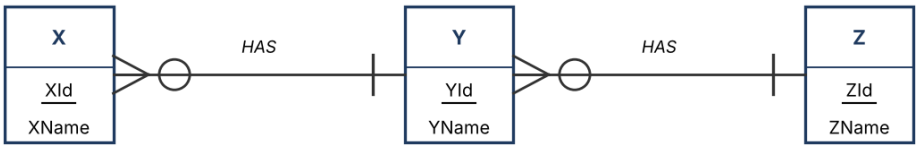
Model answer

```
STUDENT (StuId, FirstName, Surname)
    PK (StuId)
SUBJECT (SubCode, Title)
    PK (SubCode)
ENROLMENT (SubCode, StuId, EnrolDate, FinalGrade)
    PK (SubCode, StuId, EnrolDate)
    FK (SubCode) references SUBJECT
    FK (StuId) references STUDENT
```

NOTE
(a) A relational table cannot hold two values in one cell, and FK on either side would repeat or need lists. The M:M becomes a weak (intersection) entity with two M:1 links, both relationship names underlined.

Q76. [Distinction] Week 8 – trace the identifier

Read ERD e3 below (no underlined relationships). What are the primary keys and foreign keys of X, Y and Z?

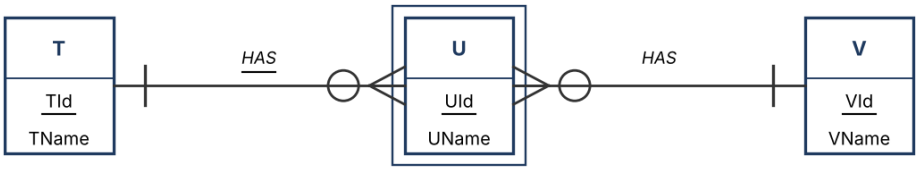


Model answer

```
Z (ZId, ZName)          PK (ZId)
Y (YId, YName, ZId)     PK (YId)    FK (ZId) → Z
X (XId, XName, YId)     PK (XId)    FK (YId) → Y
```

Q77. [Distinction] Week 8 – identifier chain

In the ERD below only the T–U relationship is underlined. Write the PK of U and V and the FKs.



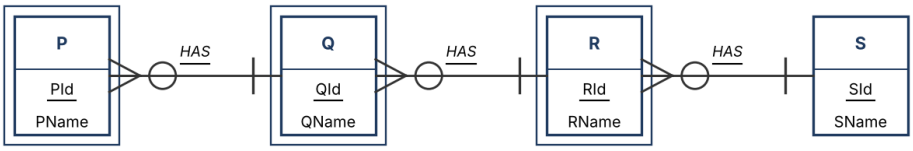
Model answer

```
T (TId, TName)              PK (TId)
U (TId, UId, UName)         PK (TId, UId)     FK (TId) → T
V (VId, VName, TId, UId)    PK (VId)          FK (TId, UId) → U
```

> **NOTE**
> U borrows T's identifier; V's relationship is not underlined so V keeps its own PK and the composite PK of U becomes a composite FK.

Q78. [Distinction] Week 8 – identifier chain

In the ERD below (e5) all three relationships are underlined. Write the PK of P, Q, R and S.



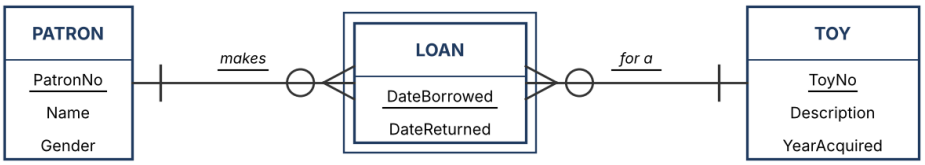
Model answer

```
S (SId, SName)                PK (SId)
R (SId, RId, RName)           PK (SId, RId)
Q (SId, RId, QId, QName)      PK (SId, RId, QId)
P (SId, RId, QId, PId, PName) PK (SId, RId, QId, PId)
```

NOTE
Each weak entity inherits the entire identifier of its parent, so keys grow down the chain.

Q79. [Distinction] Week 8 – design

Toy library: patrons (PatronNo, name, gender) borrow toys (ToyNo, description, year acquired). A loan records date borrowed and date returned (unknown until returned). The same patron can borrow the same toy more than once on different dates. Give the identifier of LOAN, justify it with the sample data, and write the schema.



Model answer

```
PATRON (PatronNo, Firstname, Surname, Gender)   PK (PatronNo)
TOY (ToyNo, Description, YearAcquired)           PK (ToyNo)
LOAN (PatronNo, ToyNo, DateBorrowed, DateReturned)
    PK (PatronNo, ToyNo, DateBorrowed)
    FK (PatronNo) → PATRON   FK (ToyNo) → TOY
```

NOTE
DateReturned can be NULL (e.g. patron 53 / toy 27 on 02-Aug-2024), and a PK can never contain NULL. (PatronNo, ToyNo) alone repeats (53, 27 three times).

Q80. [Distinction] Week 8 – constraint design

Which of these CHECK constraints are legal in Oracle? (a) CHECK (age >= 0) (b) CHECK (hiredate <= SYSDATE) (c) CHECK (salary > (SELECT AVG(salary) FROM employee)) (d) CHECK (end_date > start_date)

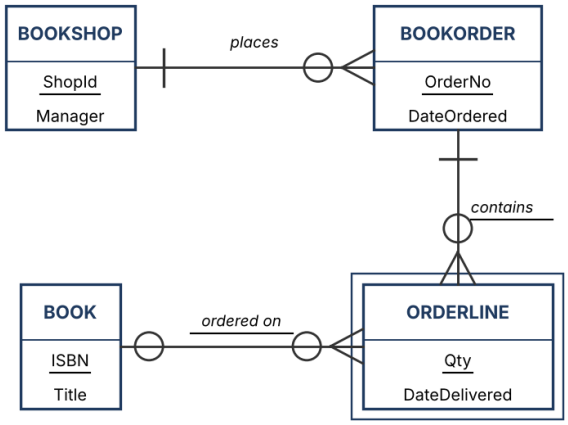
Model answer

(a) legal. (b) illegal – SYSDATE is not allowed in a CHECK. (c) illegal – subqueries are not allowed. (d) legal – compares columns of the SAME row.

Section 4 – High Distinction (HD)

Q81. [High Distinction] Integrative – ERD design

A bookshop chain: each BOOKSHOP (ShopId, Manager) places many orders; an ORDER (OrderNo, DateOrdered) belongs to exactly one shop. Each order contains many order lines, one per BOOK (ISBN, Title) ordered; a line records Qty and DateDelivered (NULL until delivered). A book may be ordered on many orders. Draw the ERD in your head (it is shown below), identify weak entities and write the schema with PKs and FKs.



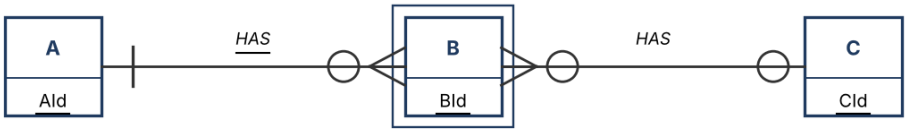
Model answer

```
BOOKSHOP (ShopId, Manager)            PK (ShopId)
BOOKORDER (OrderNo, DateOrdered, ShopId)   PK (OrderNo)   FK (ShopId) → BOOKSHOP  NOT NULL
BOOK (ISBN, Title)                     PK (ISBN)
ORDERLINE (OrderNo, ISBN, Qty, DateDelivered)
    PK (OrderNo, ISBN)
    FK (OrderNo) → BOOKORDER   FK (ISBN) → BOOK
```

> **NOTE**
> ORDERLINE is the weak (intersection) entity: both relationships are underlined, so its identifier is the two borrowed identifiers. DateDelivered may be NULL, so it cannot be in the key.

Q82. [High Distinction] Integrative – ERD critique

The ERD below (A–B and C–B) has underlined relationships on both links. Explain why this is dangerous if C is meant to be a strong entity with its own identifier, and write the schema for both interpretations.



Model answer

```
As drawn (both underlined): B (AId, CId, BId, …) PK (AId, CId, BId)  FK (AId)→A  FK (CId)→C -- B depends
    on both.
If the C–B relationship were NOT underlined: B (AId, BId, CId …) PK (AId, BId)  FK (AId)→A   FK (CId)→C
    (CId an ordinary, possibly NULL, FK).
```

NOTE
Every underline adds a borrowed identifier; extra underlines make a longer composite PK, force NOT NULL on the FK, and are marked wrong when the business rule does not require the dependency.

Q83. [High Distinction] Integrative – SQL + join + aggregate

For each campaign title, show the number of players and the number of FEMALE players. Campaigns with no players must still appear.

Model answer

```
SELECT c.campaigntitle, COUNT(p.pid) AS players,
       SUM(CASE WHEN p.gender = 'F' THEN 1 ELSE 0 END) AS female
FROM   campaign c LEFT OUTER JOIN player p ON c.campaigncode = p.campaigncode
GROUP BY c.campaigntitle;
```

Verified result: 4 row(s)

CampaignTitle	players	female
Campaign Four	0	0
Campaign One	2	1
Campaign Three	1	0
Campaign Two	2	1

NOTE
Campaign Four has 0 players; Campaign One has 2 players (1 F); Two has 2 (1 F); Three has 1 (0 F).

Q84. [High Distinction] Integrative – SQL

Toy library. List each patron's full name and the number of loans that are still OUTSTANDING (DateReturned IS NULL), including patrons with none. Sort by count descending.

Model answer

```
SELECT p.firstname || ' ' || p.surname AS patron,
       COUNT(l.toyno) AS outstanding
FROM   patron p LEFT OUTER JOIN loan l
       ON p.patronno = l.patronno AND l.datereturned IS NULL
GROUP BY p.firstname, p.surname
ORDER BY outstanding DESC;
```

Verified result: 3 row(s)

patron	outstanding
Sofia Vergara	2
Alex Dyson	0
Kim Lee	0

NOTE
Sofia Vergara 2, Alex Dyson 0, Kim Lee 0. The "outstanding" test must be in ON, not WHERE, or patrons with none would disappear.

Q85. [High Distinction] Integrative – SQL

Toys that have NEVER been borrowed by patron 71 – give two different SQL answers (NOT IN and outer join).

Model answer

```
SELECT description FROM toy
WHERE  toyno NOT IN (SELECT toyno FROM loan WHERE patronno = 71);

SELECT t.description
FROM   toy t LEFT OUTER JOIN loan l ON t.toyno = l.toyno AND l.patronno = 71
WHERE  l.toyno IS NULL;
```

Verified result (statement 1): 1 row(s)

Description
Building Blocks

Verified result (statement 2): 1 row(s)

Description
Building Blocks

> **NOTE**
> Building Blocks (never borrowed by anyone).

Q86. [High Distinction] Integrative – SQL

Students (STUDENT) with their highest-level result: list every student's name and the number of subjects they have taken; include students with no enrolments (Ravi). Then list the students who have at least one HD.

Model answer

```
SELECT s.firstname, COUNT(e.subcode) AS n
FROM   student s LEFT OUTER JOIN enrolment e ON s.stuid = e.stuid
GROUP BY s.firstname;

SELECT DISTINCT s.firstname
FROM   student s, enrolment e
WHERE  s.stuid = e.stuid AND e.finalgrade = 'HD';
```

Verified result (statement 1): 4 row(s)

FirstName	n
Helen	2
Jane	2
Jobe	2
Ravi	0

Verified result (statement 2): 2 row(s)

FirstName
Jane
Helen

Q87. [High Distinction] Integrative – predict

Without running them, decide which statements raise an Oracle error and give the code: (1) SELECT deptno, COUNT(*) FROM employee; (2) SELECT deptno, COUNT(*) FROM employee GROUP BY deptno ORDER BY 2; (3) SELECT empname FROM employee WHERE salary > AVG(salary); (4) DELETE FROM dept WHERE deptno = 10; (assuming FK from employee) (5) INSERT INTO employee (empno, empname) VALUES (8, NULL); (empname NOT NULL).

Model answer

(1) ORA-00937 not a single-group group function. (2) valid. (3) ORA-00934 group function is not allowed here. (4) ORA-02292 child record found. (5) ORA-01400 cannot insert NULL.

Q88. [High Distinction] Integrative – mock Test-2 style

Fill in the blanks: A foreign key is placed on the _____ side of a one-to-many relationship; a _____ entity's identifier includes the identifier of its parent; a value of NULL means _____; the clause that filters groups is _____; a join without a join condition produces a _____ product.

Model answer

MANY; weak; unknown / not applicable (not zero, not blank); HAVING; Cartesian.

Q89. [High Distinction] Integrative – combine code segments

Arrange these fragments into one valid Oracle statement that lists the number of movies per rating for movies longer than 120 minutes where the rating has at least 30 such movies: (i) HAVING COUNT(*) >= 30 (ii) SELECT ratingcode, COUNT(*) (iii) GROUP BY ratingcode (iv) WHERE runtime > 120 (v) FROM movie

Model answer

```
SELECT ratingcode, COUNT(*)
FROM   movie
WHERE  runtime > 120
GROUP BY ratingcode
HAVING COUNT(*) >= 30;
```

Verified result: 3 row(s)

RATINGCODE	COUNT(*)
M	93
MA	30
PG	46

> **NOTE**
> Order: SELECT → FROM → WHERE → GROUP BY → HAVING (→ ORDER BY).

Q90. [High Distinction] Integrative – design from scenario

An online course site stores Learners (LearnerNo, Name), Courses (CourseCode, Title) and Certificates. A learner can complete a course several times but never twice on the same day; the completion date and score are stored. Produce the ERD description, the identifier with justification, and the CREATE TABLE statements.

Model answer

```
CREATE TABLE Learner (LearnerNo NUMBER PRIMARY KEY, Name VARCHAR(50) NOT NULL);
CREATE TABLE Course  (CourseCode VARCHAR(10) PRIMARY KEY, Title VARCHAR(100) NOT NULL);
CREATE TABLE Completion (
  LearnerNo NUMBER, CourseCode VARCHAR(10), CompletionDate DATE, Score NUMBER CHECK (Score BETWEEN 0 AND
      100),
  PRIMARY KEY (LearnerNo, CourseCode, CompletionDate),
  FOREIGN KEY (LearnerNo) REFERENCES Learner,
  FOREIGN KEY (CourseCode) REFERENCES Course );
```

> **NOTE**
> M:M Learner–Course expanded into the weak entity COMPLETION; CompletionDate is part of the identifier because repeats are allowed on different days. Score is not in the key and may be NULL.

Section 5 – Mock Test 3 (30 minutes)

Answers for the mock test are the model answers of the same question numbers above: Q10, Q33, Q41, Q50, Q56, Q60, Q63, Q68, Q77, Q84, Q87, Q89.

Mock 1 (Q10) [Pass]

Which titles match the Access criterion Like "the ????" ?
a) The Fall b) The Matrix c) Thelma and Louise d) the Hulk e) The Ring

Answer

a), d) and e). “the ” (with the space) followed by exactly four characters; Access is not case sensitive. b) has 6 characters after “The ”; c) has no space after “The”.

Mock 2 (Q33) [Pass]

Use the CUSTOMER table (custno, custname, typecode, sales, location) supplied with the Test-2 style data (see Section 0.1). How many rows does each statement return?

Answer

Mock 3 (Q41) [Pass]

Match the Oracle error to the situation: (1) inserting a subject whose LecId does not exist in LECTURER; (2) deleting a lecturer who still convenes subjects; (3) inserting NULL into a NOT NULL column; (4) smart quotes; (5) SELECT relyear, COUNT(*) FROM movie;

Answer

(1) ORA-02291 parent key not found. (2) ORA-02292 child record found. (3) ORA-01400 cannot insert NULL. (4) ORA-00911 invalid character. (5) ORA-00937 not a single-group group function.

Mock 4 (Q50) [Credit]

List the departments that have more than one employee, with the employee count.

Answer

```
SELECT deptno, COUNT(*) AS cnt
FROM   employee
GROUP BY deptno
HAVING COUNT(*) > 1;
```

Mock 5 (Q56) [Credit]

Table T has UNIQUE(Code) and CHECK(Price > 0). For each insert say whether it succeeds: (a) (1,'A1', 5) (b) (2,'A1', 9) (c) (3, NULL, 7) (d) (4, NULL, 8) (e) (5,'B2', NULL) (f) (6,'C3', 0). Name the error codes.

Answer

(a) succeeds. (b) fails ORA-00001 unique constraint (duplicate A1). (c) succeeds – UNIQUE allows NULL. (d) succeeds – a second NULL does not count as duplicate. (e) succeeds – a CHECK passes when the condition is TRUE or NULL/UNKNOWN. (f) fails ORA-02290 check constraint violated (0 > 0 is FALSE).

Mock 6 (Q60) [Credit]

In the ERD below the relationship between E and F is underlined (E has a double outline). State the identifier of E in the relational schema and write the schema.



Answer

```
F (FId, FName)
    PK (FId)
E (FId, EId, EName)
    PK (FId, EId)
    FK (FId) references F
```

Mock 7 (Q63) [Distinction]

Departments with NO employees. (a) Write it with NOT IN. (b) Explain what happens if the subquery returns a NULL (Gina has NULL DeptNo). (c) Give a safe version.

Answer

```
SELECT deptname FROM dept
WHERE  deptno NOT IN (SELECT deptno FROM employee);

SELECT deptname FROM dept
WHERE  deptno NOT IN (SELECT deptno FROM employee WHERE deptno IS NOT NULL);

SELECT d.deptname FROM dept d
WHERE  NOT EXISTS (SELECT 1 FROM employee e WHERE e.deptno = d.deptno);
```

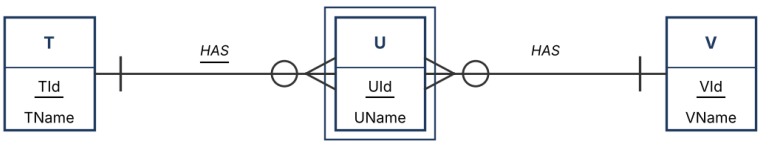
Mock 8 (Q68) [Distinction]

Table UNI_STUDENT has Amanda Allen, Bob Black, Carol Chen, Dave Dole (4 rows) and UNI_TUTOR has Carol Chen, Ed Ellis, Fred Foley (3 rows). How many rows does each return: UNION, UNION ALL, INTERSECT, MINUS (student MINUS tutor)?

Answer

Mock 9 (Q77) [Distinction]

In the ERD below only the T–U relationship is underlined. Write the PK of U and V and the FKs.



Answer

```
T (TId, TName)              PK (TId)
U (TId, UId, UName)         PK (TId, UId)     FK (TId) → T
V (VId, VName, TId, UId)    PK (VId)          FK (TId, UId) → U
```

Mock 10 (Q84) [High Distinction]

Toy library. List each patron's full name and the number of loans that are still OUTSTANDING (DateReturned IS NULL), including patrons with none. Sort by count descending.

Answer

```
SELECT p.firstname || ' ' || p.surname AS patron,
       COUNT(l.toyno) AS outstanding
```

```
FROM   patron p LEFT OUTER JOIN loan l
       ON p.patronno = l.patronno AND l.datereturned IS NULL
GROUP BY p.firstname, p.surname
ORDER BY outstanding DESC;
```

Mock 11 (Q87) [High Distinction]

Without running them, decide which statements raise an Oracle error and give the code: (1) SELECT deptno, COUNT(*) FROM employee; (2) SELECT deptno, COUNT(*) FROM employee GROUP BY deptno ORDER BY 2; (3) SELECT empname FROM employee WHERE salary > AVG(salary); (4) DELETE FROM dept WHERE deptno = 10; (assuming FK from employee) (5) INSERT INTO employee (empno, empname) VALUES (8, NULL); (empname NOT NULL).

Answer

(1) ORA-00937 not a single-group group function. (2) valid. (3) ORA-00934 group function is not allowed here. (4) ORA-02292 child record found. (5) ORA-01400 cannot insert NULL.

Mock 12 (Q89) [High Distinction]

Arrange these fragments into one valid Oracle statement that lists the number of movies per rating for movies longer than 120 minutes where the rating has at least 30 such movies: (i) HAVING COUNT(*) >= 30 (ii) SELECT ratingcode, COUNT(*) (iii) GROUP BY ratingcode (iv) WHERE runtime > 120 (v) FROM movie

Answer

```
SELECT ratingcode, COUNT(*)
FROM   movie
WHERE  runtime > 120
GROUP BY ratingcode
HAVING COUNT(*) >= 30;
```